

## Exam 1 Guide

See the Topics Schedule and Assessment sections in the Syllabus for exam schedules and procedures.

### Topics

#### 1. Basics

- **Types of differential equations.** Ordinary and partial
- **ODE classification.** Order, autonomy, homogeneity, and linearity
- **1st-order ODE forms.** Separable and 1st-order linear forms
- **Types of solutions.** Exact, general, specific, and approximate
- **Verifying solutions.** 1st-order ODEs

#### 2. Modelling Physical Systems

- **Interpreting 1st-order ODEs.** Making meaningful 1st-order ODEs
- **Population dynamics.** Exponential and logistic growth/decay
- **Newton's Law of Cooling.** Temperature difference of object and environment
- **Mixing.** Flow change with constant and non-constant volumes

#### 3. Qualitative Analysis of 1st-Order ODEs

- **Existence and uniqueness.** Initial value problems and equilibriums
- **Equilibrium solutions and stability.** Stable (attractor), unstable (repellor), semi-stable
- **Slope fields.** Slope vectors representing the rate of change on an  $t$ - $y$  plane
- **Phase diagrams.** A  $y$ - $y'$  plane representing the solution behaviors

#### 4. Solving 1st-Order ODEs

- **Analytical.** Separation of variables and integrating factors
- **Graphical.** Solving for the nullclines and isoclines
- **Numerical.** Euler's method

#### 5. Bifurcations Analysis of 1st-Order ODEs

- **One dimensional bifurcations.** Qualitative changes of solutions as a parameter changes.